

Performance Testing of Dog Breed Identification Using Transfer Learning

1. Purpose of Performance Testing

Performance testing evaluates how accurately and efficiently the trained transfer learning model identifies dog breeds from images. It ensures the model is reliable and effective on unseen data.

2. Dataset Splitting

The dataset is divided into training (70%), validation (15%), and testing (15%). The test dataset is used for final performance evaluation.

3. Performance Metrics

Accuracy measures overall correctness. Precision measures correctness of positive predictions. Recall measures how many actual breeds were correctly identified. F1 Score balances precision and recall. Loss measures model error.

4. Confusion Matrix

A confusion matrix shows correct and incorrect predictions for each breed and helps identify classification errors.

5. Training and Testing Performance

Transfer learning models such as VGG16, ResNet50, and MobileNet provide high accuracy. Example: ResNet50 achieved 91% test accuracy.

6. Prediction Speed

Performance testing also measures prediction time. Faster models like MobileNet provide quick results suitable for real-time applications.

7. Final Results

Example results: Test Accuracy: 91%, Precision: 90%, Recall: 89%, F1 Score: 89.5%, Loss: 0.23.

8. Conclusion

Transfer learning improves dog breed identification performance, reduces training time, and provides accurate classification suitable for real-world applications.

Example Model Performance Table

Model	Training Accuracy	Validation Accuracy	Test Accuracy
VGG16	92%	89%	88%
ResNet50	95%	92%	91%
MobileNet	93%	90%	89%